Summary in Graph

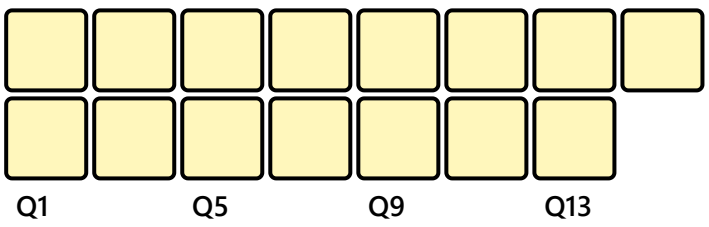
# Exam Summary (GO Classes Test Series 2026 | Discrete Mathematics | Topic Wise Test 5)

| | |
|---|---|
| Qs. Attempted: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) |
| Correct Marks: | 0 (0 + 0) |
| Penalty Marks: | 0 (0 + 0) |
| Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 15 (5 + 10) |
| Total Marks: | 25 (5 + 20) |
| Exam Duration: | 45 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (102) out of 1022 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

What is the smallest value of $n$ such that the complement of a cycle graph on $n$ vertices has an Eulerian circuit (and is connected)?

Your Answer: | Correct Answer: 5 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #2** Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

Let $K_{A,B}$ be the complete bipartite graph with $A, B$ being the two parts, and let $|A| = 10, |B| = 17$. Let $u, v \in A$ be two different vertices. We obtain a new graph $G$ by adding an edge between $u, v$. The chromatic number of $G$ is _____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #3** Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

A certain tree of order $40$ (i.e. tree with $40$ vertices) contains only vertices of degree $1$ and degree $3$. How many degree-$3$ vertices does the tree have?

Your Answer: | Correct Answer: 19 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #4** Numerical Type | Award: 1 | Penalty: 0 | Graph Theory

$G(V, E)$ is a simple undirected graph with $8$ vertices. The edges of $G$ are decided by tossing a fair coin for each two vertex combination. Edge is added between any two vertices iff head is turned up. Expected number of edges in the graph $G(V, E)$?

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #5** Multiple Select Type | Award: 1 | Penalty: 0 | Graph Theory

In each of the following problems information about the degree sequence of a graph is given. In which case, a simple undirected graph satisfying the specified conditions exists?

A. A simple graph $Q$ with degree sequence $(1, 1, 2, 3, 3, 5)$
B. A simple graph $Q$ with degree sequence $(3, 3, 3, 3)$
C. A simple graph $Q$ with degree sequence $(1, 2, 3, 4, 5, 7)$
D. A simple graph $Q$ with degree sequence $(3, 3, 3, 5)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

**Q #6** Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

A tree is a connected undirected graph with no cycles. How many non-isomorphic trees with $5$ vertices exist?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #7** Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

Let $G$ be a graph on $10$ vertices. We delete one vertex from $G$. Since we have $10$ vertices, hence we get $10$ different subgraphs depending on which vertex we have deleted. Suppose that the number of edges in the vertex-deleted subgraphs of graph $G$ are

$$12, 12, 12, 12, 11, 11, 11, 11, 10, 10.$$

How many edges are there in $G$?

Your Answer: | Correct Answer: 14 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

Let $n = 100$. What is the least number of edges possible in a connected non-planar simple graph which has $n$ vertices?

Your Answer: | Correct Answer: 103 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Consider the following properties of an undirected simple graph $G$ with $n$ vertices.

a. $G$ is connected.
b. $G$ is acyclic.
c. $G$ has $n - 1$ edges.
d. Every two different vertices in $G$ are connected by a unique path.

Which of the following is correct?

A. if $G$ satisfies properties (a) and (b), then $G$ satisfies property (c)
B. if $G$ satisfies properties (a) and (c), then $G$ satisfies property (b)
C. if $G$ satisfies properties (b) and (c), then $G$ satisfies property (a)
D. if $G$ satisfies properties (d), then $G$ satisfies property (a), (b), (c)

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Numerical Type | Award: 2 | Penalty: 0 | Graph Theory

A simple graph is an undirected graph in which each edge connects two different vertices and no two edges connect the same pair of vertices. An "oriented simple graph" is a simple graph which has been converted to a digraph by assigning an orientation to each edge. The orientation of $\{u, v\}$ can be thought of as a mapping of it to either $(u, v)$ or $(v, u)$ but not to both. Note that if $u$ and $v$ are different vertices then the orientation of $\{u, v\}$ is a mapping of it to either $(u, v)$ or $(v, u)$ but not to both. So, the following digraph is Not a oriented simple graph : $V = \{u, v\}$ and $E = \{(u, v), (v, u)\}$ What is the number of oriented simple graphs on $5$ vertices?

Your Answer: | Correct Answer: 59049 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Let $G$ be a simple graph, then $G'$ represents the complement of $G$.

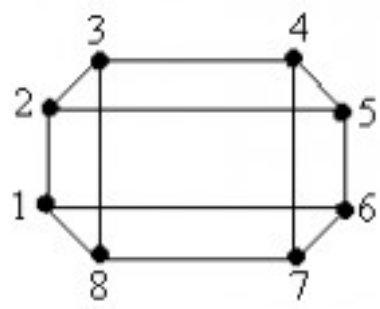
Which of the following is/are true if all the graphs that are considered are simple graphs?

A. The size of the maximum independent set in $G$ is same as the size of maximum clique on $G'$
B. The size of minimum vertex cover in $G$ is the same as the size of maximum clique on $G'$( complement of $G$)
C. The size of minimum vertex cover in $G$ is the same as the size of maximum clique on $G$.
D. The size of the maximum independent set in $G$ is same as the size of maximum clique on $G$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Consider the following graph $\mathrm{G}$ :



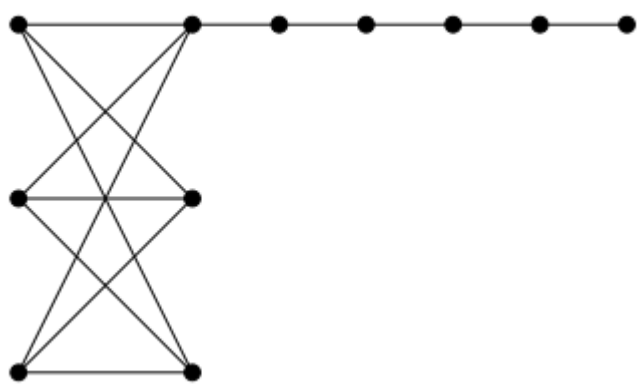

Which of the following is/are true for this graph $\mathrm{G}$?

A. Matching number of $\mathrm{G} = 4$
B. Chromatic number of $\mathrm{G} = 2$
C. $\mathrm{G}$ is a planar graph
D. $\mathrm{G}$ is a bipartite graph.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Graph Theory

Consider the following graph $\mathrm{G}$ :



Which of the following are true for this graph $\mathrm{G}$?

A. $\mathrm{G}$ is bipartite.
B. $\mathrm{G}$ is planar
C. Diameter of $\mathrm{G}$ is $6$.
D. $\mathrm{G}$ is $3$-colorable.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

Let $\mathrm{T}$ denote a nonempty binary tree in which every node either is a leaf or has two children. Then $n(\mathrm{T})$ denotes the number of non-leaf nodes of $\mathrm{T}$ (where $n(\mathrm{T}) = 0$, if $\mathrm{T}$ is a leaf), $h(\mathrm{T})$ denotes the height of $\mathrm{T}$ (where $h(\mathrm{T}) = 0$, if $\mathrm{T}$ is a leaf), $\mathrm{T_L}$ denotes the left subtree of $\mathrm{T}$, and $\mathrm{T_R}$ denotes the right subtree of $\mathrm{T}$. If $\mathrm{F}$ is a function defined by

$$\mathrm{F(T)} = \begin{cases} 0 \ \text{ if T is a leaf} \\ \mathrm{F}\left(\mathrm{T_L}\right) + \mathrm{F}\left(\mathrm{T_R}\right) + \min\left(h\left(\mathrm{T_L}\right), h\left(\mathrm{T_R}\right)\right) \ \text{otherwise,} \end{cases}$$

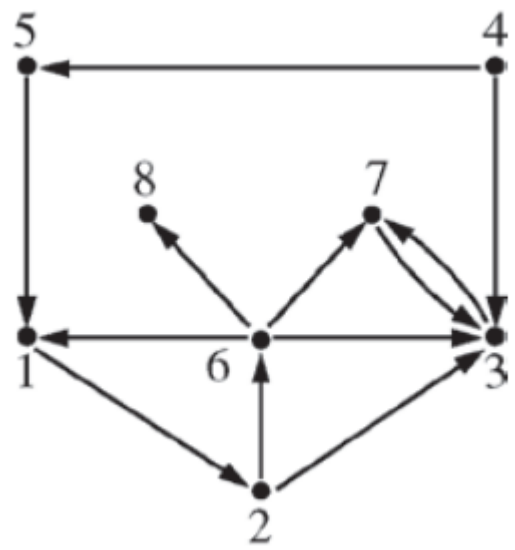
Then $\mathrm{F(T)} =$

A. $n(\mathrm{T}) + h(\mathrm{T}) - 1$
B. $n(\mathrm{T}) + h(\mathrm{T})$
C. $n(\mathrm{T}) - h(\mathrm{T}) - 1$
D. $n(\mathrm{T}) - h(\mathrm{T})$

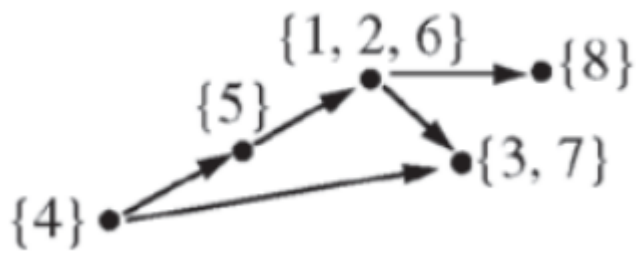
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

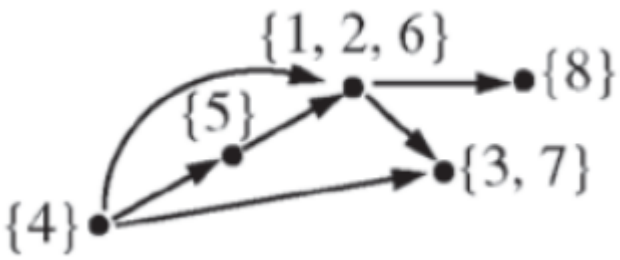
Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Graph Theory

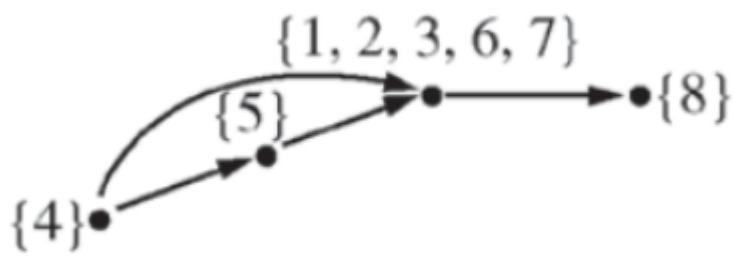
Given a directed graph $\mathrm{G} = (\mathrm{V},\mathrm{E})$, it is convenient to represent the connectivity properties of $\mathrm{G}$ using an associated directed acyclic graph $\mathrm{G}' = (\mathrm{V}', \mathrm{E}')$, where the vertices in $\mathrm{V}'$ are the strongly connected components of $\mathrm{G}$ and for $\mathrm{S}, \mathrm{T} \in \mathrm{V}'$, directed edge $(\mathrm{S},\mathrm{T})$ is in $\mathrm{E}'$ if and only if there exist $u \in \mathrm{S}$ and $v \in \mathrm{T}$ such that $(u, v) \in \mathrm{E}$. Let $\mathrm{G}$ be the graph shown below.

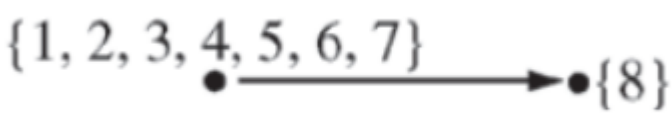



Which of the following is its associated directed acyclic graph $\mathrm{G}'$?

A.



B.



C.



D.



Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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